

Chapter 5: Parametric Regression Models

Modelling Survival Data in Medical Research

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Distributions for survival data

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Why Parametric Models?

The Cox model's strength:

The Cox proportional hazards model leaves the baseline hazard $h_0(t)$ completely unspecified. This gives it flexibility and wide applicability, no distributional assumption is needed.

The cost of that flexibility:

If a specific distribution *is* valid for the data, then inferences that assume it will be **more precise**. Estimates of relative hazards and median survival times tend to have **smaller standard errors**.

Parametric models

A model in which a specific probability distribution is assumed for the survival times.

The central distribution: Weibull

The Weibull distribution is as central to parametric survival analysis as the normal distribution is to linear modelling.

Two families of regression model:

- ▶ **Proportional hazards** (Weibull, Gompertz)
- ▶ **Accelerated failure time**, a wider family, useful when proportional hazards fails

Models for the Hazard Function

Specifying a distribution two ways:

(1) Via the density $f(t)$. Then the survivor and hazard functions follow:

$$S(t) = 1 - \int_0^t f(u) du$$

$$h(t) = \frac{f(t)}{S(t)} = -\frac{d}{dt} \{\log S(t)\}$$

(2) Via the hazard $h(t)$. Then the survivor and density functions follow from the integrated hazard.

Key relationships

$$S(t) = \exp\{-H(t)\}$$

$$f(t) = h(t) S(t) = -\frac{dS(t)}{dt}$$

where the **integrated (cumulative) hazard** is

$$H(t) = \int_0^t h(u) du$$

Specify any one of f , S , h , or H , and the other three are determined. Parametric models often start from a convenient form for $h(t)$.

The Exponential Distribution

The simplest model: constant hazard.

The risk of the event is the same at every instant, regardless of elapsed time:

$$h(t) = \lambda, \quad 0 \leq t < \infty$$

where $\lambda > 0$.

Survivor and density functions:

$$S(t) = e^{-\lambda t}, \quad f(t) = \lambda e^{-\lambda t}$$

The survival time T has an exponential distribution with mean λ^{-1} .

Percentiles:

The p th percentile (value below which a fraction $p/100$ have failed):

$$t(p) = \frac{1}{\lambda} \log \left(\frac{100}{100 - p} \right)$$

Median survival time:

$$t(50) = \frac{1}{\lambda} \log 2$$

When is it appropriate?

A constant hazard is a strong assumption — rarely tenable in practice. It serves as a baseline and a building block, and is a special case of the Weibull ($\gamma = 1$).

The Weibull Distribution

A flexible generalisation of the exponential:

$$h(t) = \lambda \gamma t^{\gamma-1}, \quad 0 \leq t < \infty$$

with two parameters, both positive:

- ▶ λ : the **scale** parameter
- ▶ γ : the **shape** parameter

Survivor and density functions:

$$S(t) = \exp(-\lambda t^\gamma)$$

$$f(t) = \lambda \gamma t^{\gamma-1} \exp(-\lambda t^\gamma)$$

Denoted $W(\lambda, \gamma)$. The distribution is positively skewed.

Behaviour of the shape parameter γ :

- ▶ $\gamma = 1$: constant hazard (exponential)
- ▶ $\gamma > 1$: hazard increases monotonically
- ▶ $\gamma < 1$: hazard decreases monotonically

Percentiles (much simpler than the mean):

$$t(p) = \left\{ \frac{1}{\lambda} \log \left(\frac{100}{100 - p} \right) \right\}^{1/\gamma}$$

$$t(50) = \left\{ \frac{1}{\lambda} \log 2 \right\}^{1/\gamma}$$

The Weibull's flexible, monotonic hazard makes it the workhorse of parametric survival analysis.

The Weibull Distribution: Hazard Shapes

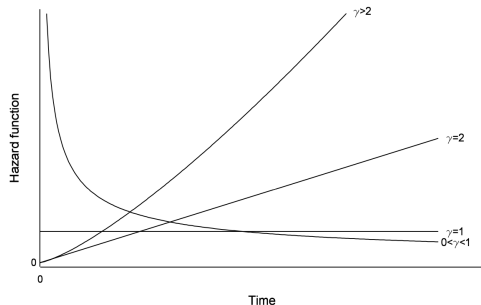


Figure: Weibull hazard function $h(t) = \lambda\gamma t^{\gamma-1}$ for different values of the shape parameter γ .

Reading the figure:

- ▶ $\gamma < 1$: the hazard starts high and *decreases* over time (e.g. early-failure “infant mortality”)
- ▶ $\gamma = 1$: the hazard is *flat* (exponential)
- ▶ $\gamma > 1$: the hazard *increases* over time (e.g. wear-out, ageing)

The shape parameter controls everything:

A single parameter γ lets the Weibull represent a wide variety of monotonic hazard behaviours, which is why it is so widely used.

Limitation: the Weibull hazard is always **monotonic**. For hazards that rise then fall, consider the log-logistic or lognormal.

The Log-logistic Distribution

For non-monotonic (unimodal) hazards:

Some hazards rise then fall. For example, after a heart transplant the risk of death rises for the first days, then falls as the patient recovers.

Hazard and survivor functions

$$h(t) = \frac{e^{\theta} \kappa t^{\kappa-1}}{1 + e^{\theta} t^{\kappa}}$$

$$S(t) = (1 + e^{\theta} t^{\kappa})^{-1}$$

with $\kappa > 0$.

Behaviour of κ :

- ▶ $\kappa \leq 1$: hazard decreases monotonically
- ▶ $\kappa > 1$: hazard rises to a single peak, then decreases (unimodal)

Percentiles:

$$t(p) = \left\{ \frac{p}{100 - p} \right\}^{1/\kappa} e^{-\theta/\kappa}$$

$$t(50) = e^{-\theta/\kappa}$$

Why “log-logistic”?

$\log T$ has a logistic distribution, symmetric, and very close in shape to the normal.

The Lognormal Distribution

Definition:

T has a lognormal distribution if $\log T$ is normally distributed with mean μ and variance σ^2 .

Density:

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} t^{-1} \exp\left\{-\frac{(\log t - \mu)^2}{2\sigma^2}\right\}$$

Survivor function (uses the standard normal CDF Φ):

$$S(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right)$$

Percentiles:

$$t(p) = \exp\{\sigma \Phi^{-1}(p/100) + \mu\}$$

$$t(50) = e^{\mu}$$

Hazard function shape:

Zero at $t = 0$, rises to a maximum, then decreases back toward zero, unimodal, like the log-logistic.

Limitation

The survivor and hazard functions can only be written in terms of integrals (no closed form). Because the normal and logistic are so similar, lognormal and log-logistic models give very similar results.

The Gompertz Distribution

Introduced for human mortality (1825).

Now used in demography and the biological sciences.

Hazard function

$$h(t) = \lambda e^{\theta t}, \quad \lambda > 0$$

- ▶ $\theta = 0$: constant hazard (exponential)
- ▶ $\theta > 0$: hazard increases with time
- ▶ The **log-hazard is linear in t** (contrast: Weibull log-hazard is linear in $\log t$)

Survivor and density functions:

$$S(t) = \exp\left\{\frac{\lambda}{\theta}(1 - e^{\theta t})\right\}$$

$$f(t) = \lambda e^{\theta t} \exp\left\{\frac{\lambda}{\theta}(1 - e^{\theta t})\right\}$$

Median survival time:

$$t(50) = \frac{1}{\theta} \log\left\{1 + \frac{\theta}{\lambda} \log 2\right\}$$

Like the Weibull, the Gompertz hazard is monotonic (increasing or decreasing), and it has the proportional hazards property.

The Gamma and Generalised Gamma Distributions

The gamma distribution:

$$f(t) = \frac{\lambda^\rho t^{\rho-1} e^{-\lambda t}}{\Gamma(\rho)}$$

with mean ρ/λ , variance ρ/λ^2 , $\rho > 0$.

- ▶ Hazard increases monotonically if $\rho > 1$, decreases if $\rho < 1$, tends to λ as $t \rightarrow \infty$
- ▶ $\rho = 1$: reduces to the exponential
- ▶ Survivor function needs the *incomplete gamma function* (no closed form)
- ▶ Quite similar to the Weibull in practice

The generalised gamma distribution:

Adds a third parameter θ , making it a flexible family that *includes several distributions as special cases*:

Special case	Condition
Weibull	$\rho = 1$
Gamma	$\theta = 1$
Lognormal	$\rho \rightarrow \infty$
Exponential	$\rho = \theta = 1$

Because it nests the others, the generalised gamma is useful for **discriminating between** alternative parametric models.

Other Distributions for Survival Data

Inverse Gaussian:

A flexible model with useful theoretical properties, but its complicated survivor function makes it difficult to work with.

Gompertz–Makeham:

A Gompertz hazard plus a constant. Models a hazard that rises (or falls) in the short term, then levels off to a horizontal asymptote:

$$h(t) = \theta - \beta e^{-\gamma t} \quad \text{or} \quad \theta + \beta e^{-\gamma t}$$

Quadratic hazard:

For a hazard that falls then rises symmetrically about a minimum:

$$h(t) = \theta + \beta t + \gamma t^2$$

(parameters chosen to keep $h(t) \geq 0$).

“Bathtub” hazard:

Falls to a single minimum then rises — the classic reliability shape (high early and late risk):

$$h(t) = \frac{\alpha t + \beta}{1 + \gamma t}$$

The choice of distribution should be guided by the expected **shape of the hazard** over time.

Log-Cumulative Hazard Plot: Construction

Idea: transform the survival function so that the result is linear if the model is correct.

Weibull model:

$$S(t) = \exp(-\lambda t^\gamma)$$

Taking logs twice:

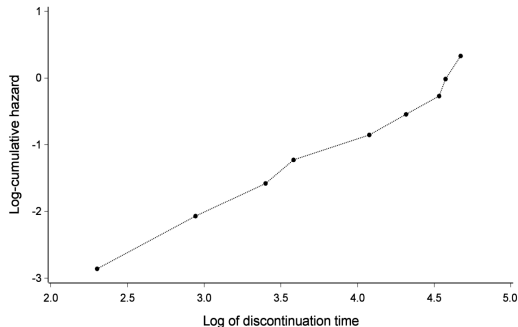
$$\log\{-\log S(t)\} = \log \lambda + \gamma \log t$$

Result: a linear function of $\log t$.

Empirical version: replace $S(t)$ with Kaplan–Meier $\hat{S}(t)$ and plot

$$\log\{-\log \hat{S}(t)\} \text{ vs } \log t$$

Log-Cumulative Hazard Plot: Interpretation



How to read the plot:

- ▶ Straight line $\Rightarrow$ Weibull model is appropriate
- ▶ Slope $\approx \gamma$ (shape)
- ▶ Intercept $\approx \log \lambda$
- ▶ Slope near 1 suggests exponential model may be sufficient

Figure: Log-cumulative hazard plot for IUD discontinuation data; linearity supports the Weibull model.

Assessing the Log-Logistic Distribution

Log-logistic model:

The log-odds of survival beyond t are linear in $\log t$:

$$\log\left(\frac{S(t)}{1 - S(t)}\right) = -\theta - \kappa \log t$$

Diagnostic plot:

- ▶ Plot estimated log-odds versus $\log t$
- ▶ Approximate linearity supports the model
- ▶ Intercept and slope estimate $-\theta$ and $-\kappa$

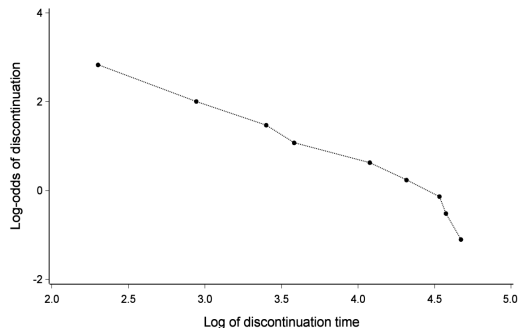


Figure: Log-odds of discontinuation versus $\log t$ for the IUD data; linearity supports the log-logistic model.

Fitting by Maximum Likelihood

No censoring:

With observed survival times $t_1, \dots, t_n$ and density $f(t)$, the likelihood is the product of the densities:

$$L = \prod_{i=1}^n f(t_i)$$

Maximise $\log L$ over the unknown parameters.

With censoring:

A death at t_i contributes $f(t_i)$. A censored observation at t_i^* contributes $S(t_i^*)$, all we know is that survival exceeded t_i^* .

The censored likelihood

With event indicator δ_i ($1 = \text{event}$, $0 = \text{censored}$):

$$L = \prod_{i=1}^n \{f(t_i)\}^{\delta_i} \{S(t_i)\}^{1-\delta_i}$$

A useful equivalent form, using $f = h \cdot S$:

$$L = \prod_{i=1}^n \{h(t_i)\}^{\delta_i} S(t_i)$$

This second form is especially convenient when the density has a complicated shape but the hazard is simple.

Fitting the Exponential Distribution

The log-likelihood:

With r deaths among n individuals, substituting $f(t) = \lambda e^{-\lambda t}$ and $S(t) = e^{-\lambda t}$:

$$\log L(\lambda) = r \log \lambda - \lambda \sum_{i=1}^n t_i$$

Maximum likelihood estimator:

Differentiate, set to zero:

$$\hat{\lambda} = \frac{r}{\sum_{i=1}^n t_i}$$

Intuitive interpretation:

$\hat{\mu} = \hat{\lambda}^{-1} = \frac{\sum t_i}{r}$ is the **total time survived divided by the number of deaths**; a natural estimate of mean lifetime from censored data.

Standard error:

From the second derivative of the log-likelihood:

$$\text{se}(\hat{\lambda}) = \hat{\lambda} / \sqrt{r}$$

The number of **events** r , not the sample size n , drives the precision of the estimate.

Exponential: Percentiles and Their Standard Errors

Estimated functions and percentiles:

$$\hat{h}(t) = \hat{\lambda}, \quad \hat{S}(t) = \exp(-\hat{\lambda}t)$$

$$\hat{t}(p) = \frac{1}{\hat{\lambda}} \log\left(\frac{100}{100-p}\right)$$

$$\hat{t}(50) = \hat{\lambda}^{-1} \log 2$$

Standard error of a percentile:

$$\text{se}\{\hat{t}(p)\} = \hat{t}(p)/\sqrt{r}$$

Confidence intervals:

Best obtained on the **log scale**, then exponentiated, this keeps the limits non-negative. Since

$$\text{se}\{\log \hat{t}(p)\} = 1/\sqrt{r},$$

the $100(1 - \alpha)\%$ limits for $t(p)$ are

$$\hat{t}(p) \exp\{\pm z_{\alpha/2}/\sqrt{r}\}$$

IUD example

With $r = 9$ events and total time 1046 days:
 $\hat{\lambda} = 0.0086$, median = 81 days, 95% CI (42, 155) days.

Fitting the Weibull Distribution

The log-likelihood (with r deaths):

$$\log L(\lambda, \gamma) = r \log(\lambda \gamma) + (\gamma - 1) \sum_i \delta_i \log t_i \\ - \lambda \sum_i t_i^\gamma$$

The score equations:

$$\hat{\lambda} = \frac{r}{\sum_i t_i^{\hat{\gamma}}} \\ \frac{r}{\hat{\gamma}} + \sum_i \delta_i \log t_i - \frac{r \sum_i t_i^{\hat{\gamma}} \log t_i}{\sum_i t_i^{\hat{\gamma}}} = 0$$

Solving the equations:

- ▶ The equation for $\hat{\gamma}$ is **nonlinear**; no closed form
- ▶ Solve numerically (Newton–Raphson)
- ▶ Once $\hat{\gamma}$ is found, $\hat{\lambda}$ follows directly
- ▶ The Newton–Raphson procedure also yields the variance–covariance matrix, hence standard errors

Estimated percentiles:

$$\hat{t}(p) = \left\{ \frac{1}{\hat{\lambda}} \log \left(\frac{100}{100 - p} \right) \right\}^{1/\hat{\gamma}}$$

In the IUD data, the Weibull median (79 days) matches the exponential (81 days), but with a **smaller standard error** (15.8 vs 26.8 days).

Weibull: Standard Error of a Percentile

The delta method for two parameters:

For a function $g(\hat{\theta}_1, \hat{\theta}_2)$ of two estimates, the approximate variance is

$$\left(\frac{\partial g}{\partial \hat{\theta}_1}\right)^2 \text{var}(\hat{\theta}_1) + \left(\frac{\partial g}{\partial \hat{\theta}_2}\right)^2 \text{var}(\hat{\theta}_2) + 2 \frac{\partial g}{\partial \hat{\theta}_1} \frac{\partial g}{\partial \hat{\theta}_2} \text{cov}(\hat{\theta}_1, \hat{\theta}_2)$$

Apply this to $\log \hat{t}(p)$, then transform back.

Working on the log scale:

Writing $c_p = \log \log\{100/(100 - p)\}$,

$$\log \hat{t}(p) = \frac{1}{\hat{\gamma}} (c_p - \log \hat{\lambda})$$

The variance of $\log \hat{t}(p)$ combines $\text{var}(\hat{\lambda})$, $\text{var}(\hat{\gamma})$, and $\text{cov}(\hat{\lambda}, \hat{\gamma})$.

Confidence interval:

Form limits for $\log t(p)$, then exponentiate:

$$\hat{t}(p) \exp\{\pm z_{\alpha/2} \text{se}(\log \hat{t}(p))\}$$

The median is estimated more precisely than other percentiles, so its CI is the narrowest.

The Weibull Model for Comparing Two Groups

A proportional hazards setup:

Let $X = 0$ for Group I, $X = 1$ for Group II. The hazard for individual i is

$$h_i(t) = e^{\beta x_i} h_0(t)$$

- ▶ Group I hazard: $h_0(t)$
- ▶ Group II hazard: $\psi h_0(t)$, where $\psi = e^\beta$
- ▶ $\beta = \log \psi$ is the log relative hazard

Now assume Group I survival times are $W(\lambda, \gamma)$, so $h_0(t) = \lambda \gamma t^{\gamma-1}$.

The proportional hazards property

If Group I is $W(\lambda, \gamma)$ and Group II has proportional hazards, then Group II is $W(\psi\lambda, \gamma)$, **same shape γ , scaled λ** .

Maximum likelihood estimates (exponential case, with r_1, r_2 deaths and T_1, T_2 total times):

$$\hat{\psi} = \frac{r_2 T_1}{r_1 T_2}, \quad \hat{\lambda} = \frac{r_1}{T_1}$$

$\hat{\psi}$ is the ratio of average times survived in the two groups, an intuitive estimate of the relative hazard.

Exploratory Analysis for Two Groups

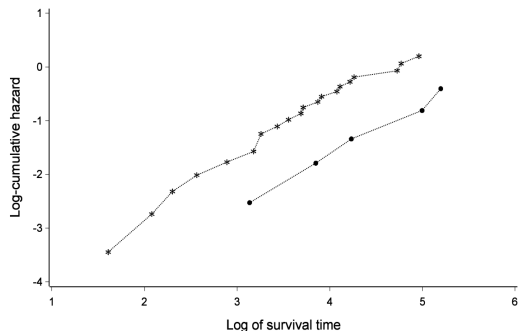


Figure: Log-cumulative hazard plot for patients with negatively stained (●) and positively stained (★) tumours. Parallel straight lines support the Weibull and proportional hazards assumptions.

What to look for:

- ▶ **Both lines straight** $\Rightarrow$ Weibull is plausible in each group
- ▶ **Lines parallel** $\Rightarrow$ proportional hazards holds (same γ)
- ▶ **Vertical separation** $\approx \beta = \log \psi$, the log relative hazard

Diagnosing departures:

- ▶ Straight but not parallel: shape γ differs between groups; hazards not proportional
- ▶ Not straight: the Weibull model may not fit

If the lines are parallel and straight, the Weibull PH model is appropriate; if parallel but curved, the Cox model may be preferable.

The Weibull Proportional Hazards Model

The general model:

With p explanatory variables, the hazard for individual i is

$$h_i(t) = \exp(\beta' x_i) \lambda \gamma t^{\gamma-1}$$

where $\beta' x_i = \beta_1 x_{1i} + \dots + \beta_p x_{pi}$.

Key difference from the Cox model:

The Cox model leaves $h_0(t)$ unspecified. Here $h_0(t) = \lambda \gamma t^{\gamma-1}$ is a **specific Weibull form**, estimated jointly with the β 's.

Survival time distribution:

Individual i has a $W(\lambda e^{\beta' x_i}, \gamma)$ distribution. The covariates **alter the scale parameter**; the shape γ stays constant.

Survivor function:

$$S_i(t) = \exp\{-\exp(\beta' x_i) \lambda t^\gamma\}$$

Estimated percentiles:

$$\hat{t}(p) = \left\{ \frac{1}{\hat{\lambda} \exp(\hat{\beta}' x_i)} \log \frac{100}{100 - p} \right\}^{1/\hat{\gamma}}$$

The model is fitted by maximising the full likelihood over $\beta_1, \dots, \beta_p, \lambda, \gamma$.

Log-linear Form of the Model

Why software uses a different parameterisation:

Most software fits the Weibull PH model in **log-linear (accelerated failure time) form**, expressing the hazard and survivor functions via parameters μ (intercept) and σ (scale):

$$S_i(t) = \exp\left\{-\exp\frac{\log t - \mu - \alpha_1 x_{1i} - \dots}{\sigma}\right\}$$

where $\alpha_1, \dots, \alpha_p$ are the coefficients in this form.

Translating between forms

$$\lambda = \exp(-\mu/\sigma), \quad \gamma = \sigma^{-1}, \quad \beta_j = -\alpha_j/\sigma$$

Recovering a hazard ratio:

Software reports $\hat{\alpha}$ and $\text{se}(\hat{\alpha})$. Then $\hat{\beta} = -\hat{\alpha}/\hat{\sigma}$, but its standard error needs the delta method:

$$\text{var}(\hat{\beta}) \approx \frac{1}{\hat{\sigma}^4} \{ \hat{\sigma}^2 \text{var}(\hat{\alpha}) + \hat{\alpha}^2 \text{var}(\hat{\sigma}) - 2\hat{\alpha}\hat{\sigma} \text{cov}(\hat{\alpha}, \hat{\sigma}) \}$$

Know your software's parameterisation, or hazard ratios will be misinterpreted.

Comparing Alternative Weibull Models

Nested model comparison:

As with the Cox model, compare nested Weibull models using the statistic $-2 \log \hat{L}$, where $\hat{L}$ is the maximised likelihood.

The likelihood ratio test:

If Model (2) adds q variables to Model (1), then under the null hypothesis that the extra coefficients are zero,

$$-2 \log \hat{L}_1 - (-2 \log \hat{L}_2) \sim \chi_q^2$$

A significantly large difference $\Rightarrow$ the extra terms are needed.

Worked example: hypernephroma

Patients classified by age group and nephrectomy status. Comparing the five candidate models by $-2 \log \hat{L}$:

Model	$-2 \log \hat{L}$
null	104.89
age	96.40
nephrectomy	94.38
age + neph.	87.76
age + neph. + interaction	83.06

The interaction reduces $-2 \log \hat{L}$ by 4.69 on 2 df ($P = 0.096$) — a borderline suggestion that nephrectomy's effect differs by age group.

The Gompertz PH Model and Model Choice

Gompertz proportional hazards model:

Since $\psi h_0(t)$ is again Gompertz when $h_0(t) = \lambda e^{\theta t}$, the Gompertz has the PH property:

$$h_i(t) = \exp(\beta' x_i) \lambda e^{\theta t}$$

Fitted by maximum likelihood; β 's are log-hazard ratios; models compared via $-2 \log \hat{L}$. Useful in demography and biological sciences, though less common than the Weibull.

Choosing between Cox and Weibull:

- ▶ Diagnostic plots (log-cumulative hazard) may be uninformative with covariates present
- ▶ Fit a Cox model and examine the shape of its baseline hazard, if smoothly increasing or decreasing, Weibull may give a more concise summary
- ▶ Compare standard errors: if the Weibull's are much smaller, prefer it for efficiency
- ▶ If similar, prefer Cox for its weaker assumptions

Discrimination is difficult unless there are many event times. Residual analysis (Chapter 7) rarely settles it decisively.

The Accelerated Failure Time Model

A different way covariates act:

Instead of multiplying the hazard, explanatory variables **act multiplicatively on the time scale**, speeding up or slowing down the passage of time.

Two-group form:

If $S_S(t)$ and $S_N(t)$ are the survivor functions for standard and new treatments,

$$S_N(t) = S_S(t/\phi)$$

The lifetime on the new treatment is ϕ times what it would have been on the standard.

Interpreting ϕ (the time ratio):

- ▶ $\phi > 1$: survival is *prolonged*
- ▶ $\phi < 1$: the event is *accelerated*
- ▶ Every percentile scales: $t_N(p) = \phi t_S(p)$

The inverse ϕ^{-1} is the **acceleration factor**.

Hazard form: writing $\phi = e^\alpha$,

$$h_i(t) = e^{-\alpha x_i} h_0(t/e^{\alpha x_i})$$

The AFT model has direct clinical appeal: it describes the **speed of disease progression**.

AFT vs. Proportional Hazards

Two ways covariates can act:

► Proportional hazards:

$h_P(t) = \psi h_0(t)$ — scales the hazard. Survivor function:

$$S_P(t) = [S_0(t)]^\psi.$$

► Accelerated failure time:

$h_A(t) = \phi^{-1} h_0(t/\phi)$ — rescales time. Survivor function:

$$S_A(t) = S_0(t/\phi).$$

For a hazard that jumps at a threshold, the AFT model shifts *when* the jump occurs; the PH model scales its *height*.

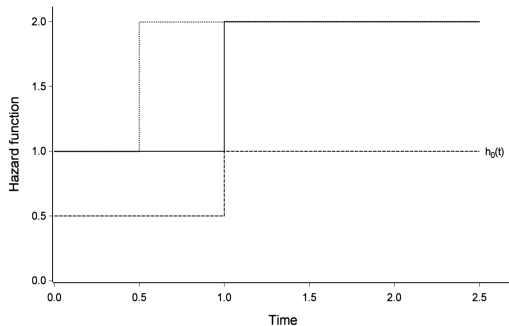


Figure: Hazard functions for Group I and II under proportional hazards (solid) and accelerated failure time (dotted) models.

A special relationship: The **Weibull is the only distribution** with *both* the proportional hazards and accelerated failure time properties. A time ratio $\phi = e^\alpha$ in AFT corresponds to a hazard ratio $e^{-\gamma\alpha}$ in PH.

The Percentile–Percentile (Q–Q) Plot

An exploratory check for the AFT model:

Estimate matching percentiles $\hat{t}_0(p)$ and $\hat{t}_1(p)$ from the two groups' Kaplan–Meier curves, for $p = 10, 20, \dots, 90$.

Under the AFT model:

$$t_0(p) = \phi^{-1} t_1(p)$$

So a plot of $\hat{t}_0(p)$ against $\hat{t}_1(p)$ should give a **straight line through the origin**, with slope ϕ^{-1} (the acceleration factor).

This is the AFT analogue of the log-cumulative hazard plot for the PH model.

Breast cancer example

The slope is roughly 3, so positive staining speeds up the disease process by a factor of about three.

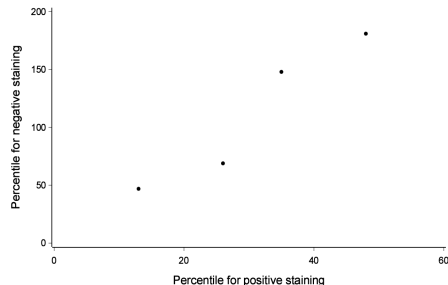


Figure: Q–Q plot for breast cancer survival times; linearity supports AFT model.

The General AFT Model and Its Log-linear Form

General hazard form:

$$h_i(t) = e^{-\eta_i} h_0(t/e^{\eta_i})$$

where $\eta_i = \alpha_1 x_{1i} + \dots + \alpha_p x_{pi}$ is the linear predictor. Survivor function:

$$S_i(t) = S_0\{t/\exp(\eta_i)\}.$$

Log-linear representation

$$\log T_i = \mu + \alpha_1 x_{1i} + \dots + \alpha_p x_{pi} + \sigma \epsilon_i$$

This looks like ordinary linear regression on $\log T$, with an error term ϵ_i whose distribution determines the survival distribution.

Interpreting the coefficients:

- ▶ Positive $\alpha_j \Rightarrow$ survival time *increases* with X_j
- ▶ The **time ratio** is $\exp(\alpha'_j x_i)$
- ▶ A one-unit increase in X_j multiplies survival time by e^{α_j}

The error distribution sets the model:

Distribution of ϵ_i	Model for T
Gumbel (extreme value)	Weibull
Logistic	Log-logistic
Normal	Lognormal

The log-linear form unifies AFT models and is what most software fits.

The Weibull and Log-logistic AFT Models

Weibull AFT model:

With baseline $h_0(t) = \lambda\gamma t^{\gamma-1}$, individual i has a $W(\lambda e^{-\gamma\eta_i}, \gamma)$ distribution. Survivor function:

$$S_i(t) = \exp\left\{-\exp\frac{\log t - \mu - \alpha'x_i}{\sigma}\right\}$$

- ▶ The error ϵ_i has a **Gumbel** distribution
- ▶ Correspondence to PH form: $\lambda = e^{-\mu/\sigma}$, $\gamma = \sigma^{-1}$, $\beta_j = -\alpha_j/\sigma$

Log-logistic AFT model:

If ϵ_i is **logistic**, T_i is log-logistic. Survivor function:

$$S_i(t) = \left\{1 + \exp\frac{\log t - \mu - \alpha'x_i}{\sigma}\right\}^{-1}$$

- ▶ Has the AFT property but *not* the PH property
- ▶ Median survival: $t_i(50) = \exp(\mu + \alpha'x_i)$ — easy to compute
- ▶ Correspondence: $\theta = -\mu/\sigma$, $\kappa = \sigma^{-1}$

The log-logistic suits unimodal hazards that the monotonic Weibull cannot capture.

The Lognormal AFT Model and Fitting

Lognormal AFT model:

If ϵ_i is **standard normal**, T_i is lognormal:

$$S_i(t) = 1 - \Phi\left(\frac{\log t - \eta_i - \mu}{\sigma}\right)$$

- ▶ Median survival: $t_i(50) = \exp(\mu + \alpha' x_i)$
- ▶ Hazard is unimodal but has no closed form

Fitting (all AFT models):

Maximum likelihood on the log-linear form. The log-likelihood is built from the survivor and density of ϵ_i , then maximised by Newton–Raphson over $\mu, \sigma, \alpha_1, \dots, \alpha_p$.

Comparing models:

Nested AFT models are compared with $-2 \log \hat{L}$, referred to χ^2 with degrees of freedom equal to the difference in the number of α -parameters.

Prostatic cancer example

Fitting a log-logistic AFT model with tumour size, index, and treatment: the estimated time ratio for treatment is $e^{0.573} = 1.77$ — treatment nearly doubles survival time, though not statistically significant ($P = 0.172$). 95% CI for the time ratio: (0.70, 4.48).

Lower $-2 \log \hat{L}$ indicates better fit; in the breast cancer data, the log-logistic (118.5) edged out the Weibull (121.8).

Measures of Explained Variation

An R^2 for parametric models:

Since the AFT model

$$\log T_i = \mu + \alpha' x_i + \sigma \epsilon_i$$

resembles a linear model, we can define an analogue of R^2 :

$$R_P^2 = \frac{\hat{V}_P}{\hat{V}_P + \hat{\sigma}^2 \text{var}(\epsilon_i)}$$

where $\hat{V}_P$ is the sample variance of the risk scores and $\hat{\sigma}^2$ is the estimated scale.

The variance term depends on the distribution:

Distribution	$\text{var}(\epsilon_i)$
Weibull	$\pi^2/6$
Log-logistic	$\pi^2/3$
Lognormal	1
Cox (set $\sigma = 1$)	—

Predictive ability:

Time-dependent ROC curves can assess predictive ability for any parametric model, once the risk score is estimated — no new principles required.

In the myeloma data, Weibull and Cox models gave very similar R^2 values, indicating comparable explanatory power.

The Proportional Odds Model

Covariates act on the odds of survival:

$$\frac{S_i(t)}{1 - S_i(t)} = e^{\eta_i} \frac{S_0(t)}{1 - S_0(t)}$$

where $\eta_i = \beta_1 x_{1i} + \dots + \beta_p x_{pi}$.

This is a linear model for the **log-odds ratio** of survival.

A distinctive property: convergent hazards.

The hazard ratio $h_i(t)/h_0(t)$ moves from $e^{-\eta_i}$ at $t = 0$ to **1** as $t \rightarrow \infty$. The hazards converge over time.

When convergent hazards make sense:

- ▶ A treatment effect that wears off over time
- ▶ A disease that is effectively cured, so groups become similar

The log-logistic proportional odds model:

The log-logistic is the natural choice; it is the **only distribution** with both the AFT and proportional odds properties. Coefficients convert via $\beta_j = \kappa \alpha_j$.

Why it is not widely used

Software for the unspecified-baseline version is limited, and it often gives results similar to a Cox model with a time-dependent covariate.

When some individuals are “cured”:

If a treatment cures a fraction of patients, the survivor function levels off above zero, the population is a **mixture** of susceptible and cured individuals.

Parametric mixture model

$$S(t) = (1 - \pi)S_n(t) + \pi$$

- ▶ π : the cured fraction (probability of cure)
- ▶ $S_n(t)$: survivor function of non-cured individuals
- ▶ As $t \rightarrow \infty$, $S(t) \rightarrow \pi$

Modelling the two pieces:

- ▶ **Non-cured survival** via a PH (or AFT) model:
 $h_{ni}(t) = \exp(\beta' x_i) h_{n0}(t)$, e.g. Weibull baseline
- ▶ **Cure probability** via logistic regression:

$$\text{logit}(\pi_i) = \phi_0 + \phi' z_i$$

The same covariates may appear in both pieces; the cure model includes an intercept ϕ_0 .

Fitting: maximum likelihood over $\beta, \phi, \lambda, \gamma$; compare models via $-2 \log \hat{L}$.

Cure-rate models separate “how likely is cure” from “how long until the event for those not cured.”

The Effect of Covariate Adjustment

A surprising contrast with linear regression:

In linear regression, adding relevant covariates *reduces* the residual variance and **increases** the precision of a treatment effect.

In survival models, this need not hold.

Adding covariates often has a negligible effect on standard errors, and the standard error of a treatment effect can even get **larger** after adjustment.

Why?

The treatment effect does not have the same interpretation in models with and without covariates (Ford, Norrie & Ahmadi, 1995). For the exponential case,

$$\text{var}(\hat{\beta}_1) = \frac{4}{n[1 - \{\text{corr}(z_1, z_2)\}^2]}$$

which is *at least* $4/n$ — the unadjusted variance.

Still include relevant covariates and check model fit: a **proper** estimate of the treatment effect requires the correctly specified model, even if precision does not improve.

- ▶ **Parametric models** assume a distribution for survival times. When correct, they yield more precise estimates than the Cox model.
- ▶ **Key distributions:** exponential (constant hazard), Weibull (monotone hazard, main workhorse), Gompertz (log-hazard linear in time), log-logistic and lognormal (unimodal hazards), and generalised gamma (flexible family that nests others).
- ▶ **Assessing fit:** log-cumulative hazard (Weibull), log-odds (log-logistic), and probit plots (lognormal). Approximate straight lines support the model.
- ▶ **Estimation:** maximum likelihood with censored likelihood $\prod \{f(t_i)\}^{\delta_i} \{S(t_i)\}^{1-\delta_i}$. Precision depends mainly on the number of events.
- ▶ **Two frameworks:** proportional hazards (Weibull, Gompertz, covariates scale hazard) and accelerated failure time (covariates rescale time). Weibull has both properties.
- ▶ **Model comparison:** based on $-2 \log \hat{L}$ and likelihood ratio tests. The log-linear form unifies AFT models and is used in software.
- ▶ **Extensions:** proportional odds models, cure rate models, and effects of covariate adjustment on precision.

Questions?

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